

✓ PDS PRACTICAL-8

Name : DHRUV GAVANKAR	Roll no.: S007
Class : SYIT	Assignment Date : 23/09/2026

✓ BeautifulSoup

1. Extract Headings Write a Python program using BeautifulSoup to extract and print all <h2> headings from a webpage.

```
import requests
from bs4 import BeautifulSoup

url = "https://www.geeksforgeeks.org/"

response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

# Get the page title
print(soup.title.text)

# Get all headings
for heading in soup.find_all("h2"):
    print(heading.text.strip())
```

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2. Extract Links Write a Python program using BeautifulSoup to extract all <a> tags and print their link text and URLs.

```
import requests
from bs4 import BeautifulSoup

url = "https://example.com"
response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")

for link in soup.find_all("a"):
```

```
print("Text:", link.get_text(strip=True))
print("URL:", link.get("href"))
```

Text: Learn more

URL: <https://iana.org/domains/example>

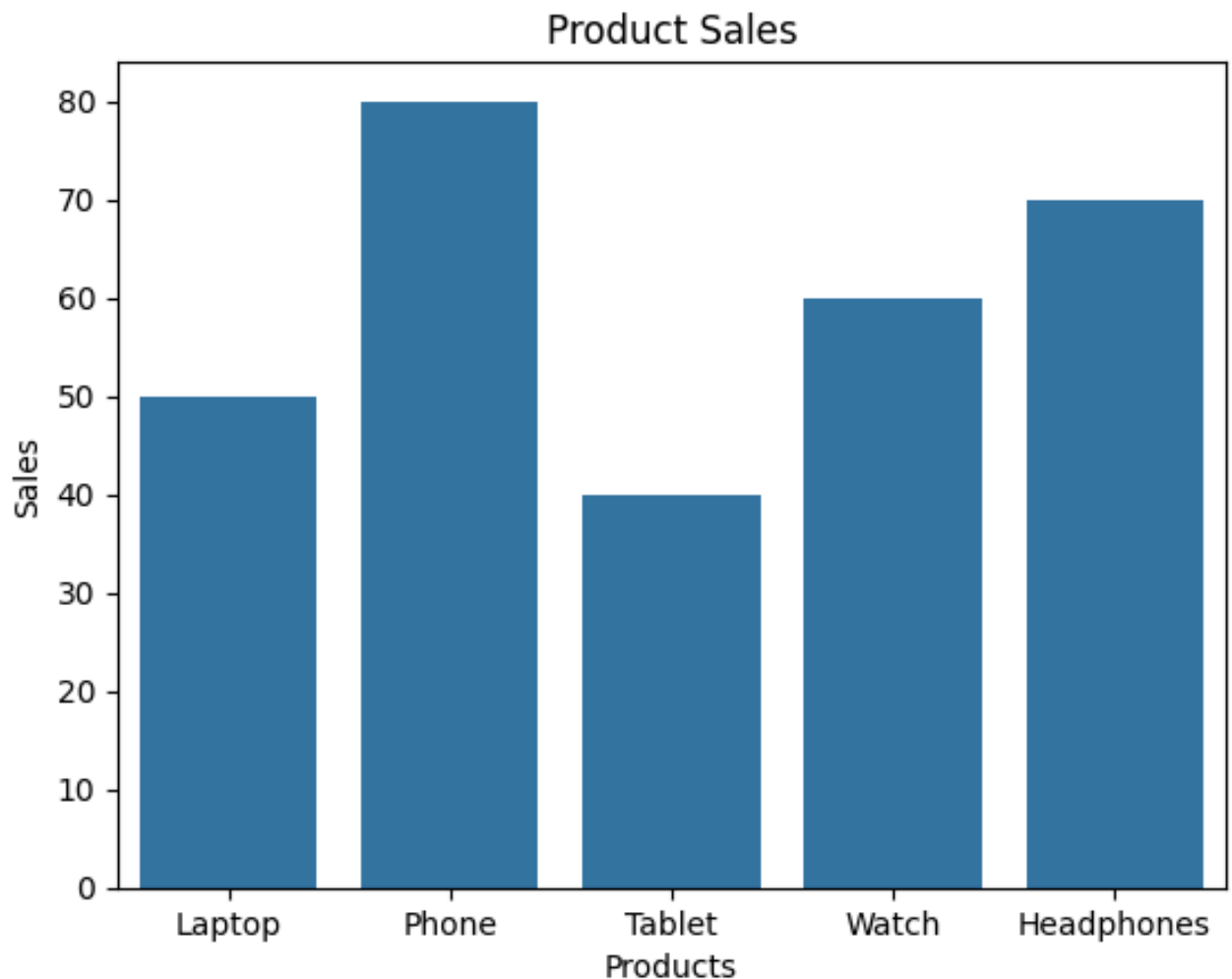
✓ Seaborn

3. Bar Chart Create a Seaborn bar chart showing the sales of 5 different products.

```
import seaborn as sns
import matplotlib.pyplot as plt

products = ["Laptop", "Phone", "Tablet", "Watch", "Headphones"]
sales = [50, 80, 40, 60, 70]

sns.barplot(x=products, y=sales)
plt.title("Product Sales")
plt.xlabel("Products")
plt.ylabel("Sales")
plt.show()
```

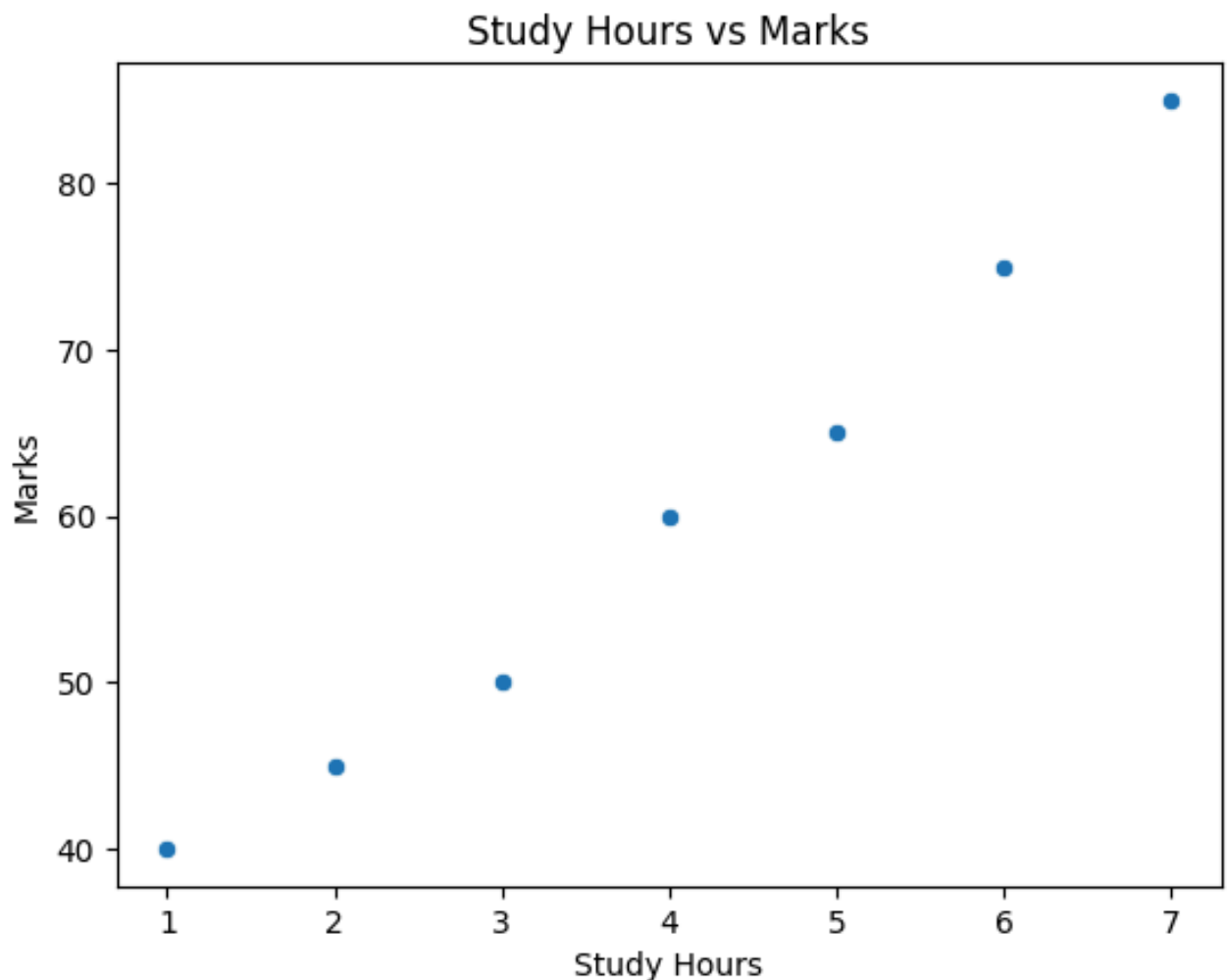


4. Scatter Plot Create a Seaborn scatter plot showing the relationship between students' study hours and their marks.

```
import seaborn as sns
import matplotlib.pyplot as plt

study_hours = [1, 2, 3, 4, 5, 6, 7]
marks = [40, 45, 50, 60, 65, 75, 85]

sns.scatterplot(x=study_hours, y=marks)
plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.show()
```



5. Histogram Create a Seaborn histogram to display the distribution of students' marks.

```
import seaborn as sns
import matplotlib.pyplot as plt

marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 85, 88, 90]
```

```
sns.histplot(marks, bins=5, kde=True)
```

```
plt.title("Distribution of Students' Marks")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

